

Lab 11

Multiple-Group Data Coding & Analysis

Instructions

In this lab, you will work with multiple datasets to practice multi-group (k-group BG ANOVA and k-group (X^2) chi-square) analyses. Make sure you have all the datasets open and review the variables before starting.

Walkthroughs (5 points)

BG Analysis – Walk-Through → Does defendant attractiveness influence recommended prison sentences?

None

A mock jury study investigated whether a defendant's physical attractiveness influences mock jurors' sentencing recommendations. In this study, 114 male participants viewed a photo of one of three female defendants who varied in attractiveness (beautiful, average attractiveness, or unattractive), then recommended a prison sentence length in years for that defendant.

The researcher hypothesized that for defendants convicted of swindling, physical attractiveness would affect sentencing severity. Specifically, they expected that beautiful defendants would receive shorter recommended sentences than both average-attractiveness defendants and unattractive defendants. Additionally, the researcher expected that average-attractiveness and unattractive defendants would receive similar sentence recommendations.

Perform a BG ANOVA to see if there are mean differences in recommended prison sentences for the different attractiveness levels.

The jury_data.sav dataset includes the following variables you will use:

- **Type of crime (crime)** - categorical/nominal; 2-levels
- **Defendant Attractiveness (attr)** - categorical/nominal; 3-levels
- **Recommended Prison Sentence (years)** - continuous scale score

Omnibus BG ANOVA

Table 1

	Beautiful	Average	Unattractive
Mean			
Std			

F = ____ df = ____ , ____ MSE = ____ p = ____ N = ____ k = ____ n = ____

LSD, Pairwise Comparisons & RH: Testing

Do we need to perform LSD pairwise comparisons to test the RH? Why or why not? _____

LSDmmd = ____

Based on the LSDmmd, use <, > & = signs to show the results of each pairwise comparison.

LSD, Pairwise Comparisons, Effect Sizes & Statistical Decision Errors

Table 2

Pairwise comparison →	Beautiful vs Average	Beautiful vs Unattractive	Average vs Unattractive
Mean difference →			
LSD result →			
Type of Stat Error risked →			
Pairwise effect size (r) →			
Power Problem? →			

Note. * No – rejecting H_0 : means there was sufficient power ** No – effect is "too small to be interesting," ($r < .10$) so being nonsignificant doesn't indicate a power problem *** Yes – The effect is "large enough to be interesting," ($r > .10$) so being nonsignificant indicates there likely is a power problem

RH: Testing

RH1: Beautiful defendants will receive shorter sentences than average-attractiveness defendants.

Is this RH: fully, partially or not supported? Explain your answer.

RH2: Beautiful defendants will receive shorter sentences than unattractive defendants.

Is this RH: fully, partially or not supported? Explain your answer.

RH3: Average-attractiveness and unattractive defendants will receive similar sentences.

Is this RH: fully, partially or not supported? Explain your answer.

Overall, is this set of RH: fully, partially or not supported? _____

Following the example in the SPSS handout, write-up the results of the omnibus-F and the pairwise comparisons. Provide a table or a figure for the data and properly refer to that table or figure in the write-up.

APA Write-up:

Homework Checker Screenshot

Paste Homework Checker screenshot here:

Chi-Square χ^2 Analysis – Walk-Through → Is defendant attractiveness associated with whether the crime is viewed as serious?

None

Using the same mock jury dataset, researchers also examined whether a defendant's physical attractiveness influenced how serious the crime was perceived to be. Participants rated the seriousness of the crime on a 1–10 scale. These ratings were then categorized as either 'Serious Crime' (ratings below 5) or 'Not Serious Crime' (ratings above 5). Ratings at the midpoint (5) were excluded. As before, only participants assigned to the swindle crime condition (crime = 2) were included.

The researcher hypothesized that physical attractiveness would be associated with perceived seriousness of the crime. Specifically, they expected that a smaller proportion of crimes by beautiful defendants would be viewed as serious compared to crimes by average-attractiveness defendants, and compared to crimes by unattractive defendants. They also expected that average-attractiveness and unattractive defendants would have similar proportions of their crimes viewed as serious.

Before performing the analysis, you will need to compute a new dichotomous variable (serious_cat) from the serious variable: Serious Crime = 1 (ratings of 1–4), Not Serious Crime = 2 (ratings of 6–10). Cases with a rating of 5 will be excluded automatically.

Perform a Pearson's χ^2 using the two qualitative variables.

The jury_data.sav dataset includes the following variables you will use:

- **Defendant Attractiveness (attr)** - categorical/nominal; 3-levels (Beautiful, Average, Unattractive)
- **Crime Seriousness (serious_cat)** - categorical/nominal; 2-levels (Serious Crime vs. Not Serious Crime)

Omnibus χ^2

Table 3

Crime Seriousness	Beautiful	Average	Unattractive
Serious			
Not Serious			

$$\chi^2 = ______ \text{ df} = ______ \text{ p} = ______ \text{ N} = ______ \text{ k} = ______ \text{ n} = ______$$

Pairwise χ^2 -critical, Pairwise Comparisons, Effect Sizes & Statistical Decision Errors

Do we need to perform pairwise χ^2 comparisons to test the RH? Why or why not? _____

$$\chi^2\text{-critical} = ______$$

Pairwise χ^2 -critical, Pairwise Comparisons, Effect Sizes & Statistical Decision Errors

Table 4

Pairwise comparison →	Beautiful vs Average	Beautiful vs Unattractive	Average vs Unattractive
% Serious →			
χ^2 result →			
Type of Stat Error risked →			
Pairwise effect size (r) →			

Pairwise comparison →	Beautiful vs Average	Beautiful vs Unattractive	Average vs Unattractive
Power Problem? →			

Note. * No – rejecting H_0 : means there was sufficient power ** No – effect is "too small to be interesting," ($r < .10$) so being nonsignificant doesn't indicate a power problem *** Yes – The effect is "large enough to be interesting," ($r > .10$) so being nonsignificant indicates there likely is a power problem

RH: Testing

RH1: A smaller proportion of crimes by beautiful defendants will be viewed as serious compared to crimes by average-attractiveness defendants.

Is this RH: fully, partially or not supported? Explain your answer.

(answer)

RH2: A smaller proportion of crimes by beautiful defendants will be viewed as serious compared to crimes by unattractive defendants.

Is this RH: fully, partially or not supported? Explain your answer.

(answer)

RH3: Average-attractiveness and unattractive defendants will have similar proportions of their crimes viewed as serious.

Is this RH: fully, partially or not supported? Explain your answer.

(answer)

Overall, is this set of RH: fully, partially or not supported?

(answer)

Following the example in the SPSS handout, write-up the results of the omnibus- X^2 and the pairwise comparisons. Provide a table or a figure for the data and properly refer to that table or figure in the write-up.

APA Write-up:

Homework Checker Screenshot

Paste Homework Checker screenshot here:

Your Turn (40 points)

BG Analysis – Your Turn → Does collegiate football experience and concussion history influence brain mass? (20 points)

None

A study examined whether football experience and concussion history affect hippocampal volume (measured in cubic centimeters). Three groups were compared: healthy control participants with no football experience or concussion history, NCAA Division 1 football players with no concussion history, and NCAA Division 1 football players with a concussion history.

The researcher hypothesized that football players with concussion history would have smaller hippocampal volumes than both football players without concussion history and control participants. Additionally, the researcher expected that football players without concussion history and control participants would have similar hippocampal volumes.

Perform a BG ANOVA to see if there are mean differences in hippocampal volume across the three groups.

The football_data.sav dataset includes the following variables you will use:

- **Football and concussion experience (group)** - categorical/nominal; 3-levels
- **Hippocampal Volume (volume)** - continuous scale score

Omnibus BG ANOVA

Table 5

	Control	FB No Con	FB Con
Mean			
Std			

F = ____ df = ____ , ____ MSE = ____ p = ____ N = ____ k = ____ n = ____

LSD, Pairwise Comparisons & RH: Testing

Do we need to perform LSD pairwise comparisons to test the RH? Why or why not? _____

LSDmmd = ____

Based on the LSDmmd, use <, > & = signs to show the results of each pairwise comparison.

LSD, Pairwise Comparisons, Effect Sizes & Statistical Decision Errors

Table 6

Pairwise comparison →	Control vs FB No Con	Control vs FB Con	FB No Con vs FB Con
Mean difference →			
LSD result →			
Type of Stat Error risked →			
Pairwise effect size (r) →			
Power Problem? →			

Note. * No – rejecting H₀: means there was sufficient power ** No – effect is "too small to be interesting," ($r < .10$) so being nonsignificant doesn't indicate a power problem *** Yes – The effect is "large enough to be interesting," ($r > .10$) so being nonsignificant indicates there likely is a power problem

RH: Testing

RH1: Football players with concussions will have lower average hippocampal volume than the control group.

Is this RH: fully, partially or not supported? Explain your answer.

RH2: Football players with concussions will have lower average hippocampal volume than football players with no concussions.

Is this RH: fully, partially or not supported? Explain your answer.

RH3: Football players without concussion history and control participants will have similar hippocampal volumes.

Is this RH: fully, partially or not supported? Explain your answer.

Overall, is this set of RH: fully, partially or not supported? _____

Following the example in the SPSS handout, write-up the results of the omnibus-F and the pairwise comparisons. Provide a table or a figure for the data and properly refer to that table or figure in the write-up.

APA Write-up:

Homework Checker Screenshot

Paste Homework Checker screenshot here:

Chi-Square χ^2 Analysis – Your Turn → Does providing customers with humorous jokes affect tipping behavior? (20 points)

None

A study was conducted to examine whether adding humor to a customer's bill affects tipping behavior at a coffee bar. When customers received their bills, they were randomly assigned to one of three conditions: the bill included a card with a joke, the bill included a card with an advertisement, or the bill had no card attached. The researcher then recorded whether each customer left a tip.

The researcher hypothesized that humor would increase tipping rates. Specifically, customers receiving joke cards would tip more frequently than those receiving advertisement cards or no card. Additionally, the researcher expected that advertisement cards would be no more effective than receiving no card at all—that is, tipping rates would be equivalent for these two conditions.

Perform a Pearson's χ^2 using the two qualitative variables.

The tips_data.sav dataset includes the following variables you will use:

- **Type of Card (card)** - categorical/nominal; 3-levels
- **Tipping Behavior (tip)** - categorical/nominal; 2-levels

Omnibus χ^2

Table 7

Tipping Behavior	Advertisement	Joke	No Card
No Tip			
Left Tip			

$\chi^2 = ______ df = ______ p = ______ N = ______ k = ______ n = ______$

Pairwise χ^2 -critical, Pairwise Comparisons, Effect Sizes & Statistical Decision Errors

Do we need to perform pairwise χ^2 comparisons to test the RH? Why or why not? $______$

χ^2 -critical = $______$

Pairwise χ^2 -critical, Pairwise Comparisons, Effect Sizes & Statistical Decision Errors

Table 8

Pairwise comparison →	Advertisement vs Joke	Advertisement vs No Card	Joke vs No Card
% Left Tip →			
χ^2 result →			
Type of Stat Error risked →			
Pairwise effect size (r) →			
Power Problem? →			

Note. * No – rejecting H_0 : means there was sufficient power ** No – effect is "too small to be interesting," ($r < .10$) so being nonsignificant doesn't indicate a power problem *** Yes – The effect is "large enough to be interesting," ($r > .10$) so being nonsignificant indicates there likely is a power problem

RH: Testing

RH1: A higher percentage of customers receiving joke cards will leave a tip compared to those receiving advertisement cards.

Is this RH: fully, partially or not supported? Explain your answer.

(answer)

RH2: A higher percentage of customers receiving joke cards will leave a tip compared to those receiving no card.

Is this RH: fully, partially or not supported? Explain your answer.

(answer)

RH3: Customers receiving advertisement cards and no card will have equal tipping rates.

Is this RH: fully, partially or not supported? Explain your answer.

(answer)

Overall, is this set of RH: fully, partially or not supported?

(answer)

Following the example in the SPSS handout, write-up the results of the omnibus- χ^2 and the pairwise comparisons. Provide a table or a figure for the data and properly refer to that table or figure in the write-up.

APA Write-up:

Homework Checker Screenshot

Paste Homework Checker screenshot here: